

Ryan Li

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TECHNICAL SKILLS

Languages: Python, C/C++, Java, R, SQL, JavaScript, TypeScript, Racket

Libraries/Frameworks: NumPy, SciPy, pandas, Plotly, scikit-learn, PyTorch, FastAPI, OpenCV, Express.js, React

Platforms/Tools: Git, Linux (Bash/Zsh), Docker, PostgreSQL, VS Code, Jupyter, Excel

HONORS & AWARDS

- **Waterloo National Mathematics Scholarship (\$15,000)**, Waterloo President's Scholarship of Distinction (\$2,000)
- **Governor General's Medal** (graduated 1/400+), Churchill Scholar's Award (\$1,250), BC Achievement (\$1,250)
- International Baccalaureate Diploma 44/45 (**top ~0.5% worldwide**)
- **Canadian Mathematics Olympiad Qualifier, Fermat Student Champion (placed 1/19644)**, 3× American Invitational Mathematics Examination Qualifier, Putnam Top 25%

EXPERIENCE

Quantitative Developer Intern

Dec 2025 – Present

Wat Street - University of Waterloo

- Engineered a 5-Minute Momentum Flag System within a 5-member subteam.
- Developed Python scripts to fetch Yahoo Finance market data and identify momentum continuation signals on 5-minute bars across **12+** equity tickers, achieving **62%** signal accuracy on backtested historical data.

Data Analyst Intern

Jun 2024 – Aug 2024

University of British Columbia

- Standardized test datasets across **12** experimental trials for cross-platform analysis, reducing manual formatting time by **40%** for researchers at UBC's Timber Engineering and Applied Mechanics Lab.
- Developed Python scripts using pandas and matplotlib automating experimental result visualization. Cut reporting turnaround by **30%** and recognized by supervisors as *"exceptionally reliable and proactive"*.

PROJECTS

Slotify (UofTHacks Winner) 🏆 | Python, JavaScript, Node.js, React, ElevenLabs, OpenAI

Jan 2026

- Built a B2B web application and API for natural insertion of sponsor reads into podcasts and audiobooks, integrating OpenAI API for contextual ad generation and ElevenLabs API for voice cloning.
- **Reduced processing latency by 35%** by supplementing LLM-based loudness and rhythm normalization with FFmpeg/librosa/pydub pipelines. Won Best Use of ElevenLabs at UofTHacks 13, placing **1/100+** teams.

Weather-Based Music Classifier 🏆 | Python, JavaScript, scikit-learn, React, Spotify Web API

Dec 2025

- Trained scikit-learn models (Logistic Regression, Gradient Boosting) for weather-based song classification, achieving **0.76 weighted F1** with 5-fold stratified cross-validation; served through a React frontend with Spotify OAuth.
- Engineered features using permutation feature importance and Spearman correlation analysis, reducing hardest class confusion (rainy/snowy) by **59%** through model iteration.

Steel Supply Chain Optimization System 🏆 | Python, JavaScript, Next.js, DS.js, pandas, FastAPI

Nov 2025

- Developed a decision-support dashboard optimizing steel supply chain routing from **50+ global manufacturers** to 26 U.S. industrial hubs. **Semi-finalist at the 11th MIT Energy and Climate Hackathon.**
- Built dynamic D3.js visualizations over **3,000+** web-scraped data points, cleaned and aggregated using pandas.

2×N Tiling Solver and Analysis Library 🏆 | C++, Python, SymPy, mpmath

Mar 2025

- Authored a **19-page** combinatorics paper on ordinary generating functions for a 2×N floor tiling problem.
- Implemented bitmask DP with backtracking for **O(N)** verification of the paper's recurrence relation.
- Corroborated approximate closed-form solutions with partial fraction decomposition in SymPy, identifying **N=8** as the threshold where floating-point error (abs error: **0.572**) causes breakdown.

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Mathematics, Honours, Computer Science Major

Sep 2025 – Present

- **96.0% Cumulative Average, 97.0% Faculty Average**
- Wat Street, Data Science Club, Quant Club, Poker Club, Intramural Volleyball